

CODEALPHA MACHINE LEARNING INTERNSHIP

TASK 2 — TECHNICAL REPORT

Speech Emotion Recognition (SER)

Deep Learning & Signal Processing Methodology

*A complete walk-through of the architectural decisions,
feature engineering, and leakage prevention strategies applied
to unstructured audio data.*

Author	Your Name
Dataset	CREMA-D (7,442 audio files, 91 actors)
Model	Convolutional Neural Network (CNN)
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1. Project Overview & Problem Definition

1.1. Objective

The objective of this project is to build a Deep Learning model capable of identifying human emotion from raw speech audio. This requires transitioning from standard tabular machine learning techniques to unstructured data handling, signal processing, and neural network architectures.

1.2. Task Type: Multiclass Classification

The output target is one of six distinct emotional states: Angry (ANG), Disgust (DIS), Fear (FEA), Happy (HAP), Neutral (NEU), and Sad (SAD). This frames the problem as a **multiclass classification** task.

Unlike imbalanced datasets where F1-Score is mandated, the chosen dataset is perfectly balanced across all six classes. Therefore, **Accuracy** serves as a reliable primary metric, supported by a Confusion Matrix to diagnose specific class overlaps (e.g., confusing Sad with Neutral).

With 6 classes, a random guessing baseline would yield an accuracy of **16.6%**. Any viable Deep Learning model must significantly outperform this mathematical baseline.

2. Data Audit & Splitting Strategy

2.1. The CREMA-D Dataset

The dataset utilized is CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset), chosen for its high audio quality and demographic diversity.

Check	Finding	Status
Dataset Size	7,442 .wav files	Sufficient for Deep Learning
Class Balance	Perfectly balanced (1,271 files per emotion)	Ideal for training
Actor Count	91 unique professional actors	Highly diverse vocal profiles

2.2. The Speaker Leakage Trap

In speech processing, the most common and fatal methodological error is random splitting. If audio files are shuffled randomly into train/test sets, files from “Actor A” will appear in both sets. Neural networks are highly adept at identifying vocal tract characteristics (timbre, pitch) rather than emotional prosody. The model will simply memorize the speaker’s voice, resulting in artificially high validation scores that collapse in production.

To ensure the model learns generalized “emotion” rather than “speaker identity,” the dataset must be split by **Actor ID**. An actor present in the training set must never appear in the validation or test sets.

Decision: Stratified Group Split — 70 / 15 / 15

The 91 unique actors were partitioned into strict, non-overlapping groups:

- **Training (70%):** ≈ 64 actors ($\approx 5,154$ files)
- **Validation (15%):** ≈ 13 actors ($\approx 1,148$ files)
- **Test (15%):** ≈ 14 actors ($\approx 1,140$ files)

This guarantees the model is evaluated exclusively on *unseen voices*, proving true generalization.

3. Signal Processing & Feature Engineering

3.1. Why Not Raw Audio?

Raw `.wav` files are 1D arrays of air pressure amplitudes over time. Feeding raw amplitudes into a neural network introduces massive noise. Instead, human auditory perception is mathematically modeled using Mel-Frequency Cepstral Coefficients (MFCCs).

Decision: Extracting 2D MFCCs via Librosa

Using the `librosa` library, each audio file was subjected to the following pipeline:

1. **Resampling:** All audio was standardized to a sample rate of **22,050 Hz**.
2. **Padding/Trimming:** Audio lengths vary. To feed a CNN, input shapes must be fixed. All files were padded with silence or trimmed to exactly **3.0 seconds** (66,150 samples).
3. **MFCC Extraction:** 40 frequency bands were extracted over time, generating a 2D matrix of shape (40, 130). This effectively converts the audio into a “heatmap” image.

4. Model Selection & Bias-Variance Diagnosis

4.1. Baseline Model: MLP (1D Vectors)

Initial testing was conducted using a Multi-Layer Perceptron (MLP). To fit the MLP, the 2D MFCC matrix was collapsed into a 1D vector of length 40 by taking the **mean across the time axis**.

Diagnosis: The MLP achieved an accuracy of only **40.0%**. By averaging the MFCCs over time, the temporal dynamics of the emotion (e.g., the sudden spike of anger, or the slow drawl of sadness) were destroyed. The model exhibited **High Variance (Overfitting)**, achieving low training loss but failing to generalize to unseen actors.

4.2. Final Model: CNN (2D Matrices)

To capture the spatial and temporal patterns inherent in speech, the architecture was upgraded to a Convolutional Neural Network (CNN). By feeding the unaltered (40, 130, 1) MFCC matrix into the CNN, the network acts as an image classifier, scanning the audio heatmap for emotional patterns.

Architecture Layer	Purpose
Input(40, 130, 1)	Receives the 2D MFCC heatmap + 1 channel dimension
Conv2D (32 filters)	Extracts low-level frequency textures
MaxPooling2D	Reduces spatial dimensions, preventing overfitting
Dropout (0.3)	Crucial regularizer: Randomly disables 30% of neurons
Conv2D (64 filters)	Combines textures into high-level emotional features
Flatten + Dense(64)	Maps spatial features to emotional logic
Dense(6, softmax)	Outputs probabilities for the 6 emotion classes

5. Evaluation & Error Analysis

5.1. CNN Performance vs. Baseline

The transition from 1D MLP to 2D CNN yielded a massive performance increase, proving that temporal sequence data is vital for emotion recognition.

Metric	MLP (Baseline)	CNN (Final)
Validation Accuracy	40.0%	46.2%
Performance vs. Random (16.6%)	$\approx 2.4\times$ better	$\approx \mathbf{2.8\times}$ better

5.2. Phase: Error Analysis via Confusion Matrix

A deep dive into the CNN’s confusion matrix on the unseen test set reveals scientifically accurate constraints regarding human speech:

- **Arousal Confusion:** The model frequently confused **Fear (FEA)** with **Happy (HAP)** (55 errors). Both emotions are characterized by high acoustic arousal, high pitch, and fast speaking rates. Without visual context, audio-only discrimination of these two classes is notoriously difficult even for humans.
- **Valence Confusion:** The model confused **Sad (SAD)** with **Neutral (NEU)** (45 errors). Both feature low vocal energy, flat pitch contours, and slower tempos.

Despite these overlaps, the model demonstrated exceptional precision on **Angry (ANG)**, achieving a heavy diagonal concentration, proving successful extraction of core emotional

signals.

6. Data Leakage Compliance & Best Practices

#	Check	Status
1	Strict Group Split applied using <code>actor_id</code> to prevent speaker leakage	PASS
2	All audio standardized to 3 seconds / 22050Hz before feature extraction	PASS
3	Train, Validation, and Test sets never intermingled during MFCC extraction	PASS
4	Model validation utilized unseen actors to monitor real-world variance	PASS
5	Final locked test set evaluated only once after architecture was finalized	PASS

7. Conclusion

This project successfully engineered a Deep Learning pipeline capable of interpreting human emotion from unstructured audio data. By transitioning from a tabular perspective to spatial signal processing (2D MFCCs + CNNs), the system captured complex vocal patterns that simpler models discard.

- **Final Test Set Accuracy:** $\approx 46.2\%$ (nearly 3 times the random baseline).
- **Robustness:** The model achieved this accuracy on a strict, leak-proof **Hold-Out Test Set** consisting entirely of voices it had never processed during training.
- **Next Steps (Optimization):** Future iterations can break the 50% accuracy barrier by applying **Phase of Data Augmentation** (injecting Gaussian noise, pitch shifting, and time-stretching) exclusively on the training set to force the CNN to generalize further.